Lecture 9

Hamilton–Jacobi theory and action–angle variables

Objectives. *Lecture 8 ended with a question. A time-dependent generating function F_2 turns Hamilton's equations for H into Hamilton's equations for the new Hamiltonian $K = H + \partial F_2/\partial t$. If some F_2 made K vanish, the new coordinates would be constant and the motion would be solved. This lecture answers the question. The condition on F_2 is one first-order partial differential equation, the* Hamilton–Jacobi equation*, and a family of its solutions with n parameters and a non-singular mixed derivative gives the motions (Jacobi's theorem). Separating the variables is how such families are found, and for a central force the parameters it finds are the energy and the angular momentum. In one degree of freedom a related aim can be met on a whole region of closed orbits at once: this yields the* action *and the* angle*, in which every oscillation becomes a uniform rotation. The period is 2π times the derivative of the action with respect to energy. For the pendulum this explains why the period of Lecture 1 diverges at the separatrix while the action does not.*

By the end of this lecture you will be able to:

- Recover motions from a complete Hamilton–Jacobi integral.
- Separate the Hamilton–Jacobi equation for a central force.
- Construct action–angle coordinates on a period annulus of a system with one degree of freedom.

9.1 The Hamilton–Jacobi equation

Let $F_2(q, P, t)$ be a C^∞ function with $\det(\partial^2 F_2/\partial q\,\partial P) \neq 0$, used on a branch as in Theorem 8.12. By that theorem, a C^1 path in the branch satisfies Hamilton's equations for H if and only if its image in the new coordinates satisfies Hamilton's equations for $K = H + \partial F_2/\partial t$ (8.19). If $K \equiv 0$, these read $\dot{Q} = 0$ and $\dot{P} = 0$: each motion is a single point of the new coordinates, and the relations (8.17) that lead back to (q, p) give the motion itself.

What does $K \equiv 0$ ask of F_2? The old momentum is $p = \partial F_2/\partial q$ by (8.17), so $K = H(q, \partial F_2/\partial q, t) + \partial F_2/\partial t$. Fix P and write $S(q, t) = F_2(q, P, t)$. Then $K \equiv 0$ says that, at every (q, t) the branch reaches,

$$\frac{\partial S}{\partial t} + H\left(q, \frac{\partial S}{\partial q}, t\right) = 0. \tag{9.1}$$

This is the *Hamilton–Jacobi equation*: one first-order partial differential equation for one function $S(q, t)$ of $n + 1$ variables.

The new momentum P does not appear in (9.1); it only labels the solution. So the question of Lecture 8 asks for a family of solutions of (9.1), labelled by n parameters. § 9.2

shows that such a family gives the motions, and § 9.3 shows how to find one. The notation S reflects a relation to the action. Fix a starting point and time, and suppose the motion from there to each nearby (q,t) is unique and depends smoothly on (q,t); then its action, read as a function of (q,t), is a solution of (9.1) (Arnold, §46).

9.2 Complete integrals and Jacobi's theorem

The F_2 of § 9.1 is a family of solutions of (9.1), labelled by the new momenta P. The next definition says what kind of family is needed.

Definition 9.1 (Complete integral). Let $G \subseteq \mathbb{R}^n \times \mathbb{R}$ and $A \subseteq \mathbb{R}^n$ be open. A *complete integral* of (9.1) on $G \times A$ is a C^∞ function $S(q,\alpha,t)$ which solves (9.1) for each fixed $\alpha \in A$ and satisfies

$$\det \frac{\partial^2 S}{\partial q\, \partial \alpha} \neq 0 \qquad \text{throughout } G \times A. \tag{9.2}$$

A complete integral is not the general solution of (9.1), which depends on an arbitrary function. It is the n-parameter family needed by Jacobi's theorem. An additive constant in S is not counted among the parameters, since it changes neither (9.1) nor (9.2).

Theorem 9.2 (Jacobi). *Let S be a complete integral on $G \times A$, fix $\alpha \in A$ and $(q_*, t_*) \in G$, and put $\beta = \partial S/\partial \alpha (q_*, \alpha, t_*)$. Then*

1. *the equations*
$$\frac{\partial S}{\partial \alpha_j}(q,\alpha,t) = \beta_j \qquad (j = 1,\dots,n) \tag{9.3}$$
determine, on a neighbourhood of t_*, a unique C^∞ curve $t \mapsto q(t)$ with $q(t_*) = q_*$;
2. *with $p(t) = \partial S/\partial q\big(q(t),\alpha,t\big)$, the pair $\big(q(t),p(t)\big)$ solves Hamilton's equations for H.*

Proof. (1) Apply the implicit function theorem to $F(q,t) = \partial S/\partial\alpha(q,\alpha,t) - \beta$, which is C^∞ and has $\partial F/\partial q = \partial^2 S/\partial\alpha\,\partial q$ invertible by (9.2), and $F(q_*,t_*) = 0$.

(2) Differentiate (9.3) along the curve:

$$\sum_k \frac{\partial^2 S}{\partial \alpha_j \partial q_k}\,\dot q_k + \frac{\partial^2 S}{\partial \alpha_j \partial t} = 0. \tag{9.4}$$

Now differentiate the Hamilton–Jacobi equation itself — which holds identically in (q,α,t) — with respect to α_j:

$$\frac{\partial^2 S}{\partial t\,\partial \alpha_j} + \sum_k \frac{\partial H}{\partial p_k}\Big(q, \frac{\partial S}{\partial q}, t\Big)\frac{\partial^2 S}{\partial q_k \partial \alpha_j} = 0. \tag{9.5}$$

Subtracting (9.5) from (9.4) at $q = q(t)$ leaves

$$\sum_k \frac{\partial^2 S}{\partial \alpha_j \partial q_k}\Big(\dot q_k - \frac{\partial H}{\partial p_k}\Big) = 0 \qquad (j = 1,\dots,n), \tag{9.6}$$

and (9.2) makes the matrix invertible, so $\dot q = \partial H/\partial p$.

For the momenta, differentiate (9.1) with respect to q_j:

$$\frac{\partial^2 S}{\partial t\,\partial q_j} + \frac{\partial H}{\partial q_j} + \sum_k \frac{\partial H}{\partial p_k}\frac{\partial^2 S}{\partial q_k \partial q_j} = 0.$$

Then, using that and $\dot q_k = \partial H/\partial p_k$,

$$\dot p_j = \sum_k \frac{\partial^2 S}{\partial q_j \partial q_k}\dot q_k + \frac{\partial^2 S}{\partial q_j \partial t} = \sum_k \frac{\partial^2 S}{\partial q_j \partial q_k}\frac{\partial H}{\partial p_k} - \frac{\partial H}{\partial q_j} - \sum_k \frac{\partial H}{\partial p_k}\frac{\partial^2 S}{\partial q_k \partial q_j} = -\frac{\partial H}{\partial q_j},$$

the two sums cancelling because second derivatives of S commute. ∎

The completeness condition (9.2) has two uses. It gives the curve $q(t)$ by the implicit function theorem and allows us to cancel the matrix in (9.6). Without it both steps can fail. For $H = (p_1^2 + p_2^2)/2$, the complete integral $S = \alpha \cdot q - \tfrac{1}{2}|\alpha|^2 t$ has mixed derivative the identity matrix, and (9.3) gives $q = \beta + \alpha t$. If instead $\partial^2 S/\partial\alpha\,\partial q$ has rank $r < n$, then (9.6) is only r independent conditions on $\dot q - \partial H/\partial p$. For the same H, the solution $S = \alpha_1 q_1 - \tfrac{1}{2}\alpha_1^2 t + f(\alpha_2)$ of (9.1), with any smooth f, has rank 1. Its relations (9.3) are $q_1 - \alpha_1 t = \beta_1$ and $f'(\alpha_2) = \beta_2$: they force $\dot q_1 = \alpha_1$ and leave $q_2(t)$ undetermined.

Read in the language of § 9.1, Jacobi's theorem answers this lecture's question. A complete integral is a type-2 generating function $F_2(q, P, t) = S(q, P, t)$ with new momentum $P = \alpha$: it is C^∞, and (9.2) is the determinant condition of Proposition 8.10, so at each fixed t it generates a canonical map near each point. Its new position is $Q = \partial S/\partial\alpha = \beta$, and its new Hamiltonian is $K = H + \partial S/\partial t = 0$ by (9.1). So the $2n$ numbers (α, β) are constant along every motion that stays in the branch, and (9.3) together with $p = \partial S/\partial q$ is the change of coordinates back. Like Theorem 8.12, all of this is local: it holds on a branch, and a complete integral found on one region of (q, t) need not extend beyond it.

9.3 Time-independent Hamiltonians and separation

When $\partial_t H = 0$ one may look for a complete integral of the form

$$S(q, \alpha, t) = W(q, \alpha) - E(\alpha)\, t. \tag{9.7}$$

Then $\partial_t S = -E(\alpha)$ and (9.1) becomes the *time-independent* Hamilton–Jacobi equation

$$H\Big(q, \frac{\partial W}{\partial q}\Big) = E(\alpha), \tag{9.8}$$

while (9.2) reads $\det(\partial^2 W/\partial q\,\partial\alpha) \neq 0$. The relations (9.3) become

$$\frac{\partial W}{\partial \alpha_j}(q, \alpha) - \frac{\partial E}{\partial \alpha_j}(\alpha)\, t = \beta_j. \tag{9.9}$$

It is usually convenient to take $\alpha_1 = E$ itself, when E may serve as a parameter; the first of (9.9) is then $\partial W/\partial E - t = \beta_1$, which gives t in terms of q — the *quadrature*, an integral of a known function — while the remaining $n-1$ relations $\partial W/\partial\alpha_j = \beta_j$ describe the orbit without reference to time.

Example 9.3 (The oscillator)**.** Let $H = p^2/2m + \tfrac{1}{2}m\nu^2 q^2$ and take the single parameter $\alpha = E > 0$. On the interval $|q| < \sqrt{2\alpha/m\nu^2}$, where the radicand $2m\alpha - m^2\nu^2 q^2$ is positive, the branch $W' > 0$ of (9.8) is

$$W(q, \alpha) = \frac{q}{2}\sqrt{2m\alpha - m^2\nu^2 q^2} + \frac{\alpha}{\nu}\arcsin\Big(q\nu\sqrt{\tfrac{m}{2\alpha}}\Big). \tag{9.10}$$

Its mixed derivative is $\partial^2 W/\partial q\,\partial\alpha = m/\sqrt{2m\alpha - m^2\nu^2 q^2} = m/p$, which is positive on that interval, so $S = W - \alpha t$ is a complete integral on any product $G \times A$ of open sets on which $|q| <$

$\sqrt{2\alpha/m\nu^2}$. The α-derivative of W is $\arcsin\!\big(q\nu\sqrt{m/2\alpha}\big)/\nu$, so (9.9) reads $\arcsin\!\big(q\nu\sqrt{m/2\alpha}\big) = \nu(t+\beta)$, that is

$$q(t) = \sqrt{\frac{2\alpha}{m\nu^2}}\,\sin\!\big(\nu(t+\beta)\big) \qquad \text{for } |\nu(t+\beta)| < \frac{\pi}{2}. \tag{9.11}$$

This is a motion of energy α, on the half-oscillation where $p = m\dot q$ is positive, which is the branch that was chosen. The other half comes from the branch $W' < 0$. A complete integral describes one arc of a motion at a time.

Separation of variables

One method for finding a complete integral is to seek W as a sum of functions of individual coordinates. Whether this separation is possible depends on the coordinates. For a central force, spherical coordinates work.

Let $\mathcal{L} = \frac12 m\,|\dot{\mathbf q}|^2 - V(|\mathbf q|)$ on $\mathbb{R}^3\setminus\{0\}$, as in Proposition 3.16, in spherical coordinates (r,ϑ,φ) — radius, colatitude $0<\vartheta<\pi$, azimuth φ. Then $\mathcal{L} = \frac12 m\big(\dot r^2 + r^2\dot\vartheta^2 + r^2\sin^2\vartheta\,\dot\varphi^2\big) - V(r)$, so $p_r = m\dot r$, $p_\vartheta = mr^2\dot\vartheta$, $p_\varphi = mr^2\sin^2\vartheta\,\dot\varphi$ and

$$H = \frac{1}{2m}\Big(p_r^2 + \frac{p_\vartheta^2}{r^2} + \frac{p_\varphi^2}{r^2\sin^2\vartheta}\Big) + V(r). \tag{9.12}$$

Proposition 9.4 (Separation for a central force)**.**

1. The separated equations. *Let $W = W_r(r) + W_\vartheta(\vartheta) + W_\varphi(\varphi)$ solve $H(q,\nabla W) = E$ with H as in (9.12). Then W_φ' is a constant α_φ, and, for a constant α_ϑ,*

$$\Big(\frac{\mathrm dW_\vartheta}{\mathrm d\vartheta}\Big)^2 + \frac{\alpha_\varphi^2}{\sin^2\vartheta} = \alpha_\vartheta^2, \qquad \Big(\frac{\mathrm dW_r}{\mathrm dr}\Big)^2 = 2m\big(E - V(r)\big) - \frac{\alpha_\vartheta^2}{r^2}. \tag{9.13}$$

   Conversely, any W of that form built from (9.13) solves $H(q,\nabla W) = E$.

2. A complete integral. *On a connected patch where both* radicands

$$\alpha_\vartheta^2 - \frac{\alpha_\varphi^2}{\sin^2\vartheta} > 0 \qquad \textit{and} \qquad 2m\big(E - V(r)\big) - \frac{\alpha_\vartheta^2}{r^2} > 0 \tag{9.14}$$

   hold, with a fixed sign for each square root and a branch of the azimuth, this gives a solution by three quadratures for each $\alpha = (E,\alpha_\vartheta,\alpha_\varphi)$, and it is a complete *integral: the mixed determinant is*

$$\det \frac{\partial^2 W}{\partial(r,\vartheta,\varphi)\,\partial(E,\alpha_\vartheta,\alpha_\varphi)} = \frac{m\,\alpha_\vartheta}{p_r p_\vartheta} \neq 0. \tag{9.15}$$

3. The constants. *They are the angular momentum $\mathbf L = \mathbf q\times\mathbf p$ of Proposition 3.16: at every point $(q,\nabla W)$, $\alpha_\varphi = L_3$ and $\alpha_\vartheta^2 = |\mathbf L|^2$.*

Proof. Substituting W into (9.12) and multiplying by $2mr^2\sin^2\vartheta$ isolates $(W_\varphi')^2$ as a function of (r,ϑ) alone, so W_φ' is a constant, α_φ. Multiplying instead by $2mr^2$ turns $H(q,\nabla W) = E$ into

$$r^2\Big[(W_r')^2 - 2m\big(E - V(r)\big)\Big] + \Big[(W_\vartheta')^2 + \frac{\alpha_\varphi^2}{\sin^2\vartheta}\Big] = 0,$$

where $W_\varphi' = \alpha_\varphi$ was used. The first term depends only on r and the second only on ϑ, so the equation holds identically exactly when the second bracket is a constant α_ϑ^2 and the first is

$-\alpha_\vartheta^2/r^2$, which is (9.13). Every step reverses, which is the converse. For (9.15), differentiate $p_r = W_r'$, $p_\vartheta = W_\vartheta'$ and $p_\varphi = \alpha_\varphi$: the last row of the matrix is $(0, 0, 1)$, and the 2×2 block above it is upper triangular with diagonal entries $\partial p_r/\partial E = m/p_r$ and $\partial p_\vartheta/\partial \alpha_\vartheta = \alpha_\vartheta/p_\vartheta$.

Finally, generalised momenta are not Cartesian components, so the Cartesian momentum $\mathbf{p}$ must be recovered first. Let e_r, e_ϑ and e_φ be the orthonormal spherical directions. Then $\partial \mathbf{q}/\partial r = e_r$, $\partial \mathbf{q}/\partial \vartheta = r e_\vartheta$ and $\partial \mathbf{q}/\partial \varphi = r \sin\vartheta\, e_\varphi$, so the rule $p_j = \mathbf{p} \cdot \partial \mathbf{q}/\partial q_j$ of Proposition 4.1 gives

$$\mathbf{p} = p_r e_r + \frac{p_\vartheta}{r} e_\vartheta + \frac{p_\varphi}{r \sin\vartheta} e_\varphi, \qquad \mathbf{L} = r e_r \times \mathbf{p} = p_\vartheta e_\varphi - \frac{p_\varphi}{\sin\vartheta} e_\vartheta.$$

Since $e_3 \cdot e_\varphi = 0$ and $e_3 \cdot e_\vartheta = -\sin\vartheta$, this gives $L_3 = p_\varphi = \alpha_\varphi$. Also $|\mathbf{L}|^2 = p_\vartheta^2 + p_\varphi^2/\sin^2\vartheta$, which is α_ϑ^2 by the first equation of (9.13). ∎

So separation has found three constants of the motion, E, $|\mathbf{L}|^2$ and L_3, as the parameters of a complete integral, and by § 9.2 the parameters of a complete integral are constant along every motion it describes. Spherical coordinates separate every central force.

9.4 Action and angle in one degree of freedom

Instead of asking the new Hamiltonian to vanish, one can ask it to depend on the new momentum alone. Then the new momentum is constant and the new position moves at a constant rate. In one degree of freedom this can be done on a whole region of closed orbits at once, not only on one arc. We do not use a generating function here. We build the coordinates directly from time and energy. Throughout this section

$$H = \frac{p^2}{2m} + V(q), \qquad V \in C^\infty(\mathbb{R}), \quad m > 0,$$

which covers the oscillator and the pendulum. The *action variable* I defined below is not the action of a motion (Lecture 3), though both are traditionally called the action.

Definition 9.5 (Period annulus)**.** Let $z_* \in \mathbb{R}^2$ with $H(z_*) = E_0$. An open set A with $z_* \notin A$ is a *period annulus around* z_* *over* (E_0, E_1) if

1. $\mathrm{d}H \neq 0$ on A;
2. for each $E \in (E_0, E_1)$ the level set $\gamma_E = A \cap H^{-1}(E)$ is a single periodic orbit of X_H, and these exhaust A;
3. each γ_E is a simple closed curve whose enclosed open region D_E satisfies $D_E \subseteq A \cup \{z_*\}$ and $H < E$ on D_E.

Such regions are common. Let q_* be a critical point of V with $V''(q_*) > 0$. Then Proposition 1.15, applied to $\ddot q = -V'(q)/m$, whose energy is H/m, supplies $a < q_* < b$ and $\varepsilon > 0$ such that for $V(q_*) < E < V(q_*) + \varepsilon$ the part of the level $H = E$ over (a, b) is a single periodic orbit: the two branches $p = \pm\sqrt{2m(E - V)}$ over $[q_-(E), q_+(E)]$. These are simple closed curves, they nest, $H < E$ strictly inside each, and $\mathrm{d}H = (V', p/m)$ vanishes on none of them, so together they form a period annulus round $(q_*, 0)$.

Proposition 9.6 (Orientation, and the area)**.** *On a period annulus, γ_E traversed by the flow runs* clockwise *in the (q, p) plane, and*

$$\oint_{\gamma_E} p \,\mathrm{d}q = \operatorname{area}(D_E) > 0. \tag{9.16}$$

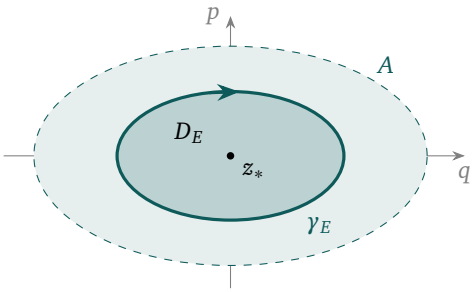


Figure 9.1: A period annulus A around z_* (light shading, without the point z_*). The level curves γ_E fill it, and each γ_E encloses a region D_E (darker) on which $H < E$. The arrow shows the sense of the flow.

Proof. On γ_E we have $H = E$ while $H < E$ on D_E, so the non-zero ∇H, which is normal to the level curve, cannot point into D_E: a step inward along it would raise H above E. It therefore points *out*. The velocity is $X_H = J\nabla H$ (6.1), and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ *is* the clockwise quarter turn, $(a, b) \mapsto (b, -a)$. So the tangent to γ_E is the outward normal turned clockwise, which is the clockwise sense of travel; this is the clockwise sense of Figure 6.1, in general form. Lecture 8 evaluated the loop integral for either orientation: by (8.12), $\oint p\,\mathrm{d}q$ is minus the enclosed area anticlockwise and plus it clockwise. ∎

Definition 9.7 (The action variable)**.** On a period annulus,

$$I(E) = \frac{1}{2\pi} \oint_{\gamma_E} p\,\mathrm{d}q = \frac{\operatorname{area}(D_E)}{2\pi}. \tag{9.17}$$

The oscillator shows the whole construction with explicit formulas. For $H = p^2/2m + \frac{1}{2}m\nu^2 q^2$, every level $E > 0$ is an ellipse round the origin, so $\mathbb{R}^2 \setminus \{0\}$ is one period annulus over $(0, \infty)$. Lecture 8 computed $\oint p\,\mathrm{d}q = 2\pi E/\nu$ (8.13), so

$$I = \frac{E}{\nu}, \qquad T = 2\pi \frac{\mathrm{d}I}{\mathrm{d}E} = \frac{2\pi}{\nu}. \tag{9.18}$$

The period does not depend on the energy, and the relation between T and $\mathrm{d}I/\mathrm{d}E$ is the one the next theorem proves in general. The right turning point is $q = \sqrt{2E/m\nu^2}$, and measuring time from it gives the angle $\varphi = \nu t$, so

$$q = \sqrt{\frac{2I}{m\nu}}\cos\varphi, \qquad p = -\sqrt{2m\nu I}\sin\varphi.$$

This map has Jacobian determinant 1, and in these coordinates $H = \nu I$ and $\dot\varphi = \nu$. The theorem builds the same coordinates for every period annulus: the angle is time rescaled, and the action is energy rescaled.

Theorem 9.8 (Action–angle in one degree of freedom · *proved modulo T1, T5 (cited)*)**.** *Let $H = p^2/2m + V(q)$, let A be a period annulus over (E_0, E_1), and let $T(E)$ be the period of γ_E. Then*

1. *T and I are C^∞, and*
$$I'(E) = \frac{T(E)}{2\pi} > 0, \tag{9.19}$$
so I is strictly increasing and has a C^∞ inverse $I \mapsto E(I)$;
2. *each γ_E has exactly one* right turning point*, a point where $p = 0$ and $V'(q) > 0$.*

Let $\varphi = 2\pi t/T(E)$, where $t \in [0, T(E))$ is the time since the motion last passed that point. Then $(\varphi, I) \mapsto (q, p)$ is canonical on each branch of the angle φ, and in these coordinates

$$H = E(I), \qquad \dot{I} = 0, \qquad \dot{\varphi} = \nu(I) := \frac{\mathrm{d}E}{\mathrm{d}I} = \frac{2\pi}{T(E)}. \tag{9.20}$$

In particular $T(E) = 2\pi\,\mathrm{d}I/\mathrm{d}E$.

Proof. The key step is (9.21): time and energy are coordinates on the annulus with Jacobian determinant 1, and the action and the angle only rescale them. The steps before it make those coordinates well defined.

The turning points. Follow the motion round γ_E. Where $p = 0$ we have $V(q) = E$, and $\dot{p} = -V'(q) \neq 0$ because $\mathrm{d}H = (V', p/m)$ does not vanish on A. So p changes sign at each of its zeros: from + to − where $V' > 0$, and from − to + where $V' < 0$. The motion is periodic, so q cannot increase or decrease for ever, and zeros of both kinds occur; they alternate. Take a zero at $q = b$ with $V'(b) > 0$ and the next one, at $q = a$. Between them $p < 0$, so $a < b$, and $V = E - p^2/2m < E$ on (a, b). After a the motion runs to the right with $p > 0$ until p next vanishes, at a point where $V = E$; the first such point to the right of a is b. So γ_E meets the axis $p = 0$ at exactly two points, and $\sigma(E) = (b, 0)$ is its only right turning point. Write $b = q_+(E)$.

A section. Fix $E' \in (E_0, E_1)$. Since $V'(q_+(E')) \neq 0$, the implicit function theorem gives a C^∞ function g near E' with $V(g(E)) = E$ and $g(E') = q_+(E')$. For E near E' the point $(g(E), 0)$ lies in the open set A and has $H = E$, so it lies on γ_E, and $V'(g(E)) > 0$ by continuity: it is $\sigma(E)$. So σ is C^∞ on (E_0, E_1), $H(\sigma(E)) = E$, and σ meets each γ_E exactly once.

The period. Let $\Phi(t, E) = \varphi_t(\sigma(E))$, which is C^∞ by Theorem 1.8, and let Φ_p be its p-component. For fixed E', the zeros of $\Phi_p(\cdot, E')$ on $[0, T(E') + \delta]$, for small $\delta > 0$, are 0, the time t_- at which the motion reaches its left turning point, and $T(E')$; each is simple, since $\partial_t \Phi_p = -V'(q) \neq 0$ there.

Away from small intervals round these three times $\left|\Phi_p(t, E')\right|$ is bounded below by a positive constant, since it is continuous and non-zero on a closed bounded set of times. By continuity the same holds for E near E'. So for such E the zeros of $\Phi_p(\cdot, E)$ on $[0, T(E') + \delta]$ are 0, one near t_- and one near $T(E')$, each unique there and the last one C^∞ in E by the implicit function theorem.

At the zero near t_- the sign of V' is still negative, and at the zero near $T(E')$ still positive. So the motion first returns to its right turning point $\sigma(E)$ at the latter, which is therefore $T(E)$, and T is C^∞.

Time and energy as coordinates. Write $w = \partial\Phi/\partial E$. Since $H \circ \Phi = E$ identically, $\nabla H \cdot w = 1$, while $\partial\Phi/\partial t = X_H$. Therefore

$$\det\left[X_H \mid w\right] = \det \begin{bmatrix} H_p & w_1 \\ -H_q & w_2 \end{bmatrix} = H_p w_2 + H_q w_1 = \nabla H \cdot w = 1. \tag{9.21}$$

By Definition 9.5, Φ maps the open set $R = \{(t, E) : E_0 < E < E_1,\ 0 < t < T(E)\}$ bijectively onto A minus the curve σ: different E give different orbits, since $H \circ \Phi = E$, and within one orbit $t \in (0, T(E))$ meets every point but $\sigma(E)$ exactly once. Its Jacobian determinant is $1 \neq 0$, so it is a diffeomorphism, and σ, a curve on the axis $p = 0$, has zero area. So for $E_0 < E < E' < E_1$ the change of variables formula gives

$$\operatorname{area}(D_{E'}) - \operatorname{area}(D_E) = \operatorname{area}\bigl(\Phi(R \cap \{E < e < E'\})\bigr) = \int_E^{E'} \int_0^{T(e)} \mathrm{d}t\,\mathrm{d}e = \int_E^{E'} T(e)\,\mathrm{d}e,$$

whence $\mathrm{d}(\text{area})/\mathrm{d}E = T(E)$ and (9.19). Since T is C^∞ and positive, so is I', and I is a C^∞ increasing bijection onto its image.

To construct the angle, set $\varphi = 2\pi t/T(E)$ on $\Phi(R)$ and $\varphi = 0$ on σ, an angle defined modulo 2π, and consider $(\varphi, I) \mapsto (t, E) = \big(\varphi T(E(I))/2\pi, E(I)\big) \mapsto \Phi(t, E)$. The first map has $\partial t/\partial\varphi = T/2\pi$, $\partial E/\partial\varphi = 0$ and $\partial E/\partial I = \mathrm{d}E/\mathrm{d}I = 2\pi/T$ by (9.19), so its Jacobian determinant is $(T/2\pi)(2\pi/T) = 1$; composing with Φ, whose determinant is 1 by (9.21), gives determinant 1. For $n = 1$ that is the condition $\{q, p\} = 1$ of Theorem 8.2, so the map is canonical. Since H is constant on each orbit and equal to $E(I)$, Hamilton's equations in the new coordinates read $\dot\varphi = \partial H/\partial I = \mathrm{d}E/\mathrm{d}I$ and $\dot I = -\partial H/\partial\varphi = 0$; and $\varphi = 2\pi t/T$ advances at exactly $2\pi/T$, as it must. ∎

For the oscillator the period does not depend on the energy. For the pendulum it does.

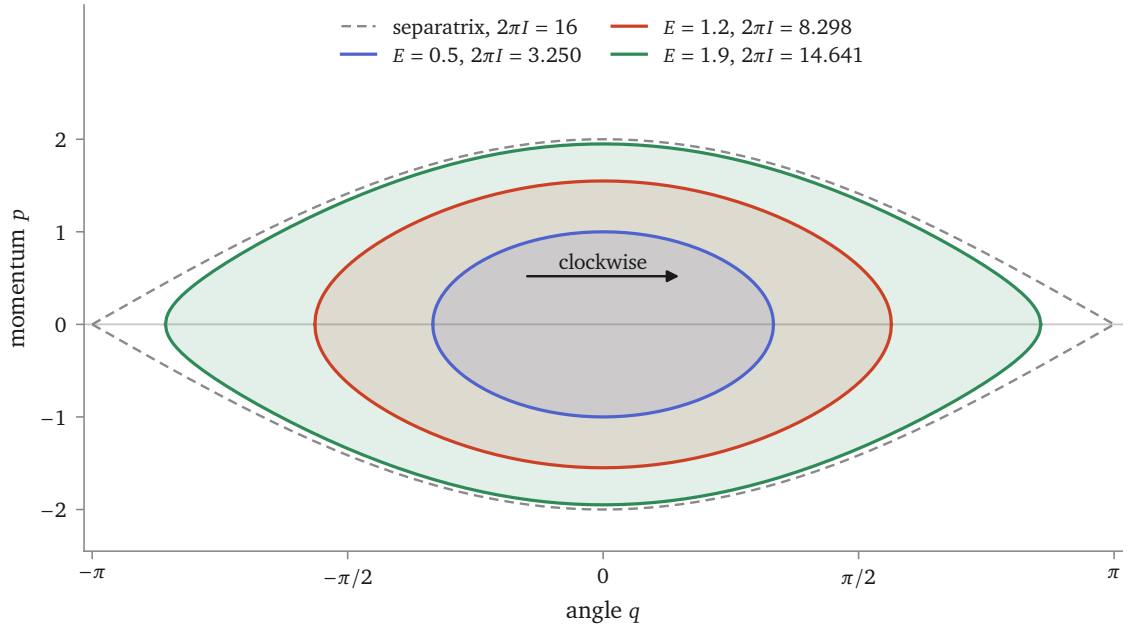


Figure 9.2: The quantity $2\pi I$ as an enclosed area, for three pendulum librations, against the separatrix. The orbits are traversed clockwise (Proposition 9.6), so $\oint p\,\mathrm{d}q$ is + the shaded area. The three numbers are computed by quadrature of the drawn curves.

Example 9.9 (The pendulum: a finite action and an infinite period). Let $H = \frac{1}{2}p^2 + 1 - \cos q$, the normalisation of Lecture 1 ((1.12): centre at $E = 0$, separatrix at $E = 2$). For $0 < E < 2$ the set $\gamma_E = \{|q| < \pi,\ H = E\}$ is the closed curve $p = \pm\sqrt{2(E - 1 + \cos q)}$ over $|q| \le q_+(E) = \arccos(1 - E)$; on it $\mathrm{d}H = (\sin q, p)$ never vanishes, since $p = 0$ only at $q = \pm q_+$, where $0 < q_+ < \pi$ and $\sin q_+ \neq 0$. The enclosed region $\{|q| < \pi,\ H < E\}$ contains the origin, and the curves nest, so the librations form a period annulus round $(0, 0)$ over $(0, 2)$, and Theorem 9.8 applies to all of it.

The separatrix $p = \pm 2\cos(q/2)$, $|q| \le \pi$, bounds the region $\{|q| < \pi,\ H < 2\}$, whose area is $\int_{-\pi}^{\pi} 4\cos(q/2)\,\mathrm{d}q = 16$, and that region is the increasing union of the D_E. So as $E \uparrow 2$ the area of D_E increases to 16, and

$$I(E) \ \uparrow\ \frac{16}{2\pi} = \frac{8}{\pi} \approx 2.5465 :$$

the action has a *finite* limit at the separatrix (Figure 9.2). The period does not: Lecture 1 found $T(E) = 4K(k)$ with $k^2 = E/2$ (1.14), which diverges logarithmically as $E \uparrow 2$ (1.15). By (9.19) that period is $2\pi\,\mathrm{d}I/\mathrm{d}E$.

What diverges at the separatrix is not the action but its slope $\mathrm{d}I/\mathrm{d}E$.

Summary.

- *The question.* An F_2 that makes the new Hamiltonian $K = H + \partial F_2/\partial t$ vanish solves the motion; the condition is the Hamilton–Jacobi equation $\partial_t S + H(q, \partial_q S, t) = 0$ (9.1).
- *The motions.* A complete integral $S(q, \alpha, t)$, with $\det(\partial^2 S/\partial q\,\partial\alpha) \neq 0$, gives motions by $\partial S/\partial\alpha = \beta$ and $p = \partial S/\partial q$ (Theorem 9.2). Read as F_2 it makes $K = 0$, and (α, β) are the constant new coordinates.
- *Separation.* For time-independent H, $S = W - Et$. The central force separates in spherical coordinates, and the constants it finds are E, $|\mathbf{L}|^2$ and L_3 (Proposition 9.4).
- *Action and angle.* On a period annulus of $H = p^2/2m + V$, the action $I = \frac{1}{2\pi}\oint p\,\mathrm{d}q$ and the angle φ are canonical coordinates in which $H = E(I)$ and the motion is a uniform rotation, with $T = 2\pi\,\mathrm{d}I/\mathrm{d}E$ (Theorem 9.8). The pendulum's action is still finite, $8/\pi$, at the separatrix, where its period diverges.
- *Next.* The constants L_3 and $|\mathbf{L}|^2$ come from the rotations, and Lecture 10 turns a group of symmetries into conserved quantities directly, without solving anything. Lecture 12 takes $I = \frac{1}{2\pi}\oint p\,\mathrm{d}q$, one closed curve at a time, as the actions of an integrable system.

References

Arnold, *Mathematical Methods of Classical Mechanics*, §46 (pp. 248–257: the action function), §47A (p. 258: $\mathbf{p}\,\mathrm{d}\mathbf{q} - \mathbf{P}\,\mathrm{d}\mathbf{Q} = \mathrm{d}S$, in our signs), §47B (pp. 260–261: Jacobi's theorem and its proof), §47C (pp. 261–265: separation of variables), and §50B (pp. 280–282: action–angle variables in one degree of freedom, with the oscillator and the pendulum as its two examples). His ω is $\mathrm{d}p \wedge \mathrm{d}q$ and §50B works in the (p, q) plane, so his orientations are not ours; only his statements are used. Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, §7.9, Theorem 7.9.3 (p. 213): Theorem 9.2 in the form "S transports X_H to equilibrium". Tong, *Lectures on Classical Dynamics*, §4.5 (pp. 105–112: action–angle variables) and §4.7 (pp. 121–125: Hamilton–Jacobi).